Kalp Shah

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Education

Indian Institute of Technology Gandhinagar **July 2024 – April 2028**

Bachelor of Technology in Electrical Engineering with Minor in Computer Science and Engineering *GPA: 8.64/10*

Relevant Coursework

Data Structures and Algorithms I, Machine Learning, Foundations of AI: Multiagent Systems, Introduction to Competitive Programming, Computing, Data-Centric Computing, Digital Systems, Probability and Statistics

Experience

Graviton Research Capital **May 2027 – November 2027**

Quantitative Developer Intern *Incoming*

AIResQ ClimSols **May 2026 – July 2026**

Machine Learning and Computer Vision Intern

- Engineered a **Computer Vision** pipeline for estimating depth of floods using CCTV Camera Networks
- Developed a **custom Camera and Radar device** for generation and validation of data in flood areas
- Implemented single view metrology algorithms for extracting features achieving **accuracy of ±3cm**

ABSix Robotics **May 2025 – July 2025**

Software Development Intern

- Engineered a **cross-platform desktop application (Python, Qt)** with real-time robot-student communication via the Dynamixel SDK deployed across **15+ challenges** and actively used by **100+ students**
- Integrated a **RAG pipeline (LangChain + ChromaDB)** into the platform to query robotics concepts

Teaching

SC 428: Introduction to C++ and the STL **October 2026 – November 2026**

Course Instructor, IIT Gandhinagar

- Co-instructing an introductory institute short course teaching **C++ programming** to beginners
- Designing lectures and practice sets on control flow, **time complexity**, arrays, vectors and the **STL**

Projects

Trade Matching Engine | *C++ 26* **Github**

- Engineered a **C++ 26** order matching engine implementing price-time priority with FIFO Order execution
- Implemented **order matching, cancellation** and **trade execution** in O(1) time using stable **std::hive iterators**
- Benchmarked the matching engine across **200K+ orders** using **hardware RDTSC timers**, achieving **12.5M+ orders/sec** throughput with **60 ns median latency**

CHIP-8 Emulator | *C++ 17* **Github**

- Built a **CHIP-8 interpreter** in with SDL2 that implements all **34 opcodes**, 4 KB of memory, a 16-level stack, and a 64×32 XOR-drawn display with keypad input and square-wave audio
- Decoupled **60 Hz timers** from a configurable CPU rate (700 instructions/sec by default) using a time accumulator, and implemented **4 runtime-configurable quirk flags** for compatibility with both COSMAC VIP and SCHIP programs

Fairness and Efficiency in Shared GPU Allocation Mechanisms | *Advisor: Prof. Manisha Padala* **Report**

- Built a custom simulator for **shared GPU clusters** to evaluate how different **scheduling mechanisms** balance allocation efficiency, long-run fairness, and resistance to strategic user manipulation
- Conducted **rigorous counterfactual simulations** using a repeated unit-demand model to **stress-test algorithms** under varying resource scarcity (10% to 80% capacity) and heterogeneous user demands
- Demonstrated that an optimized history-penalised scoring achieved **near-optimal efficiency (> 95% theoretical welfare)** and **near-perfect fairness (Jain's Index ∼0.991)**, while neutralizing malicious priority inflation

Realtime Carbon Credit Market Analysis Platform | *Python, Langchain, Next.js, PostgreSQL* **Github**

- Architected a **full-stack real-time analytics platform** serving live carbon credit pricing, trend signals, and risk analysis to end users as part of an 8-person team at **Inter-IIT Tech Meet 14.0**
- Developed a **multi-agent RAG system** generating context-aware recommendations on incoming scraped information
- Built a data streaming pipeline using **Kafka** and Debezium to capture live PostgreSQL changes, cutting data freshness latency from **60 minutes** to **under 3 seconds**

Identifying Landing Locations on Mars using Classical CV | *Advisor: Prof. Shanmuga* **Research Poster**

- Developed a CV pipeline to identify **safe landing zones on Mars** with **ISRO** under compute constraints
- Designed a modular system utilizing Sobel derivatives for **slope detection**, Fast Fourier Transforms for **texture analysis**, and the Maximal Rectangle Algorithm for **zone identification**
- Achieved a verifiable, low-power solution with **1.5 seconds runtime** on consumer-level hardware

Technical Skills

Languages: Python, C++, C, HTML/CSS, JavaScript, SQL
Frameworks: PyTorch, OpenCV, LangChain, FastAPI, Numpy, Pandas, Matplotlib, Seaborn, scikit-learn
Tools and Build Systems: Linux, Docker, Git, Github, CMake, Jupyter Notebook

Achievements

Codeforces - Handle: *kalpshah1808* — **Expert**: 1746
Finalist at AMS Derive (Rank 26/2500+) conducted by **Jane Street** and **Qube Research and Technologies**
Selected to **Barclays** BIIC Mentorship program to learn about Capital Markets and FX Hedging and ranked in Top 5
JEE Mains 2024 – All India Rank **3917** and JEE Advanced 2024 – All India Rank **6637** from 1.4 Million candidates
Integral Cup Season 1 **All India Rank 47** out of 2500+ students conducted by **Optiver** and **Da Vinci Trading**
Department Change from Mechanical to Electrical Engineering awarded based on CGPA merit after first year

Positions of Responsibility

Contingent Leader, Inter IIT Tech Meet 15.0 – Leading IIT Gandhinagar's contingent across all problem statements
Technical Coordinator, Technical Council '26–27 – Overseeing the institute's umbrella body for all technical clubs
Outreach Coordinator, Technical Council '25–26 – Managed outreach and external engagement for the Technical Council
Sponsorship Team, Amalthea '24 – Secured corporate sponsorships for IIT Gandhinagar's annual technical summit
ADH Mentor, ES115: Design, Innovation and Prototyping – Mentored first-year teams through hardware design projects
Technical Team, Entrepreneurship Cell (E-Cell) – Built and maintained technical infrastructure for E-Cell initiatives